

Relevant Papers on the Edelstein Effect in Rashba Systems

1. Edelstein Effect in Isotropic and Anisotropic Rashba Models

- **arxivID:** 2503.20712
- **Author:** Irene Gaiardoni, Mattia Trama, Alfonso Maiellaro, Claudio Guarcello, Francesco Romeo, Roberta Citro
- **date:** 2025-03-26
- **url:** <https://arxiv.org/abs/2503.20712>
- **short summary:** *Very recent.* Investigates spin-to-charge conversion via the Edelstein effect in a 2D Rashba electron gas using the semiclassical Boltzmann approach. Directly addresses calculating magnetization magnitude and direction under applied electric fields. Explores dependencies on key model parameters such as spin-orbit coupling strength, Fermi velocity, and chirality, providing explicit analytical frameworks and graphical results for both isotropic and anisotropic models.

2. Out-of-plane Edelstein effects: Electric-field induced magnetization in p -wave magnets

- **arxivID:** 2501.01888
- **Author:** Motohiko Ezawa
- **date:** 2025-01-03
- **url:** <https://arxiv.org/abs/2501.01888>
- **short summary:** *Very recent.* Analytically and numerically studies electric-field induced magnetization in p -wave magnets and Rashba systems. Details how out-of-plane and in-plane magnetization components vary with different directions and magnitudes of applied electric fields, offering clear parameter dependencies and explicit graphics for model validation.

3. Acoustic Rashba-Edelstein effect

- **arxivID:** 2107.03115
- **Author:** Takumi Funato, Mamoru Matsuo
- **date:** 2021-07-07
- **url:** <https://arxiv.org/abs/2107.03115>
- **short summary:** Theoretically studies the induction of spin density via Rashba spin-orbit interaction. Calculates linear response of spin density to external perturbations, identifying direct Edelstein and acoustic mechanisms. Provides microscopic formulas dependent on Fermi velocity and SOI strength, with graphical analysis of magnetization direction and magnitude.

4. Spin accumulation at nonmagnetic interface induced by direct Rashba Edelstein effect

- **arxivID:** 1805.05523

- **Author:** Florent Auvray, Jorge Puebla, Mingran Xu, Bivas Rana, Daisuke Hashizume, Yoshichika Otani
- **date:** 2018-05-15
- **url:** <https://arxiv.org/abs/1805.05523>
- **short summary:** Explores the direct Edelstein effect at nonmagnetic interfaces, detailing how charge currents generate spin polarization via Rashba coupling. Discusses dependencies on electric field magnitude and spin-orbit parameters, providing foundational models and explicit graphics for calculating interface magnetization.

5. Electric-Dipole Spin Resonances

- **arxivID:** 1812.01721
- **Author:** E. I. Rashba, V. I. Sheka
- **date:** 2018-11-29
- **url:** <https://arxiv.org/abs/1812.01721>
- **short summary:** Examines resonance phenomena driven by electric fields in solids with Rashba spin-orbit coupling. Offers theoretical insights into how electric field oscillations induce spin dynamics and magnetization, highlighting parameter dependencies crucial for modeling Edelstein responses at the Brillouin zone center.

6. Out-of-plane spin polarization from in-plane electric and magnetic fields

- **arxivID:** cond-mat/0609078
- **Author:** Hans-Andreas Engel, Emmanuel I. Rashba, Bertrand I. Halperin
- **date:** 2006-09-05
- **url:** <https://arxiv.org/abs/cond-mat/0609078>
- **short summary:** Demonstrates the joint effect of spin-orbit and electric/magnetic fields on perpendicular spin polarization in 2D Rashba systems. Provides baseline calculations for magnetization direction and magnitude at the Gamma point, with explicit dependence on scattering rates, Fermi velocity, and applied field strengths.